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A low-cost UWB microwave imaging system for early-stage breast cancer detection

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Abstract
In this paper, a modern and effective approach with high resilience for microwave imaging of breast tumor detection using 12 antennas is presented. The proposed antenna array operates from 3.1 GHz to 11.6 GHz according to FCC frequencies, which enables the imaging system to mix two advantages at the same time, deep penetration, and high resolution for accurate image acquisition. The suggested antenna array has a compact size of $21 \times 21 \times 12$ cm^3, this allows for the integration of as many elements as feasible, provided that the coupling between them remains within an acceptable range. This ensures improved accuracy and resolution in the resulting image. The 3D system is arranged in a circular manner around the phantom to cover all breast positions. A measurement campaign will then be made in different scenarios to detect the tumor using two artificial breasts with and without tumor, their dielectric properties and physiological arrangement approximate those of a natural breast. The proposed microwave imaging system can detect and identify all targets, making it one of the most effective systems for detecting breast tumors.

Keywords Antenna · UWB · Artificial breast · Microwave imaging · Breast cancer

1 Introduction

In recent years, there has been a noticeable surge in research dedicated to utilizing microwaves for detection and localization. Among the many examples of ongoing research, there has been increased interest in the use of microwaves for diagnostic imaging of breast

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cancer. Breast cancer is most common among the female population, with a rate at least twice that of any other tumor. Early detection significantly enhances the prospects of long-term survival for breast cancer patients, with cure rates reaching up to 90%, which certainly makes early detection a key factor in the long-term survival of breast cancer patients [1–3].

Currently, among the most frequently employed detection techniques are X-ray mammography, ultrasound, and magnetic resonance imaging (MRI) [4, 5]. Mammography is an X-ray imaging of a compressed breast, is the most used diagnostic technique to detect breast tumors. However, ionizing radiation poses a health risk, while breast compression induces considerable discomfort in patients [6, 7]. In addition, circumstances such as the presence of dense tissue around the tumor, the absence of micro-calcifications in the early stages of the disease, tumors located near the chest wall or armpits lead to a false negative rate [8, 9]. The limitations of X-ray mammography have motivated the development of imaging tools [10–13]. An ideal breast cancer screening tool should have a low health risk, be capable of detecting specific malignancies, enable early-stage detection and therefore curability, be non-invasive, cost-effective, and widely accessible, while minimizing patient discomfort and providing reliable results for interpretation [14, 15].

Currently, microwave-imaging technology (MWI) is the best method for detecting breast cancer [15–17]. This technology provides a balanced combination of resolution and depth penetration by utilizing both high and low frequencies effectively [17, 18]. Microwave imaging techniques give better results because they have the potential for a much higher sensitivity in detection and offer several advantages, including non-ionizing electromagnetic radiation, more accurate, low-cost, and comfortable without breast compression [19, 20, 43, 44].

In this paper, a new 3D system is proposed for breast cancer detection. It contains 12 antennas placed around the phantom in a circular manner. The antenna array of the imaging system is used as a sensor to send and receive microwave signals to the human body. In the beginning, we recognized that a UWB antenna presents various advantages, such as its compact size, ease of fabrication, wide bandwidth, straightforward design, and robust performance, all while maintaining compliance with FCC standards. Subsequently, two artificial breasts with and without tumors were fabricated in such a way that their dielectric properties and physiological arrangement approximated those of a natural breast. Finally, a 3D system is proposed with 12 antennas placed around the phantom in a circular manner to cover all breast positions. The targets are clearly identified and correctly located, effectively proving the performance of the manufactured system as a possible tool for breast cancer detection.

The structure of the rest of this paper is outlined as follows: Section 2 conducts a comprehensive review of the relevant literature, Section 3 introduces the designed antenna, Section 4 provides detailed insights into the breast mold design, Section 5 offers a brief overview of the system components, and Section 6 presents the results and engages in discussions. Finally, Section 7 provides a summary of our conclusions, including the 2D image of the breast tumor, and outlines potential avenues for future research.

2 Related work

In this section, we provide an overview of prior research conducted in the realm of UWB microwave imaging systems employing antennas for the early detection of breast cancer. The section also includes some existing works focused on UWB microwave imaging

system with antennas that used in medical field, also some studies using antenna for breast cancer detection. In this regard, we can find Ruvio et al. [21] created a wideband antenna operating between 1–2.5 GHz. Unfortunately, the resulting bandwidth was insufficient to meet MI's requirements for good data rate and resolution. Similar to this, Sugitani et al. [22] suggested a 4*4 array antenna to operate between 6 and 12.5 GHz. The antenna is huge, but not all of the UWB bandwidth was covered. In order to identify the breast tumor, the authors in [23] simulates a step structure wideband antenna on a breast phantom model in several positions. For the same goal, a wearable wideband patch antenna was created in [24]. The antenna was constructed on a flexible substrate to accommodate a bandwidth of 1.6 to 11.2 GHz. Although the projected antenna dimension was big, the antenna demonstrated good signal penetration for the detection of breast tumors. A UWB MIMO antenna with improved isolation was presented in [25], but the antenna was huge and lacked a directional pattern. An effective annular antenna system was proposed by A.S. Narkhede et al. [26]. Yet, because each antenna element in the system was huge due to the annular orientation, the system as a whole was massive. In contrast to FR4, a 4*4 array antenna was proposed in [27] to operate with a bandwidth of 12.5 GHz for radar-based breast cancer diagnosis. In [28], a thin-film flexible antenna operating between 2–4 GHz is suggested for the diagnosis of breast cancer. For the purpose, a ring-shaped rectangular array antenna operating in the 2–11 GHz band was suggested in [29]. To find tumors, the antenna employs a unique absorption rate approach. A bowtie antenna with a broad impedance surface that can operate between 8 and 17 GHz was also proposed in [30]. The main trade-offs associated with the designs in [27–30] include high size and limited UWB bandwidth, despite the fact that the antennas demonstrated good applicability in microwave imaging of breast cancers. Microwave imaging technology, specifically radar-based Microwave Imaging (MWI), has emerged as a prominent method for advancing biomedical detection in recent years [15, 16, 43]. This innovative technology achieves a harmonious balance between spatial resolution and depth of penetration through the utilization of both high and low frequencies. Microwave imaging techniques deliver superior outcomes, leveraging the benefits of non-ionizing electromagnetic radiation, providing a comfortable, and gentle sensing approach [19]. The process involves the transmission of electromagnetic waves through a transmitter antenna and their reception via a receiver antenna. Consequently, UWB antennas play a pivotal role in radar systems, facilitating the reduction of cluttered data to generate highly localized images [20].

Within this technological framework, a crucial focus is placed on antennas, demanding dedicated study and exploration. Various research teams are actively engaged in enhancing antenna performance to align with the latest applications [19, 20]. The development of printed antennas presents several challenges, notably the miniaturization of the radiating structure and their integration into a compact volume, all while preserving key properties, especially radiation efficiency and bandwidth [5, 6]. This technology utilizes a wide frequency band, specifically in the range from 3.1 GHz to 10.6 GHz, in line with the Federal Communications Commission (FCC) standard [5, 7].

Our proposed Ultra-Wideband (UWB) microwave imaging system represents a significant advancement in imaging technology, distinguishing itself from traditional systems in multiple crucial aspects [40, 41]. Unlike conventional methods constrained by narrow or broadband techniques, UWB technology capitalizes on a wide frequency spectrum, enabling exceptional resolution and penetration capabilities [17, 18]. By emitting short pulses with high bandwidths, UWB systems excel in discerning intricate details and precisely localizing targets, transcending the limitations of existing imaging modalities [17–20]. Additionally, their inherent resistance to interference and adaptability to various

environments highlight the versatility and robustness of UWB systems, positioning them as indispensable tools across diverse medical applications. Our hemispherical system for breast cancer imaging, employing a multistatic radar approach with an array of 12 original UWB antennas, demonstrates promising results. These antennas, strategically positioned to avoid mutual interference, ensure optimal data acquisition. Rigorous testing, based on meticulously studied distances, and experimental measurements on realistic breast phantoms, showcase the system's viability as a compelling alternative to mammography. Offering advantages such as non-ionizing radiation, affordability, comfort, and minimal side effects, our proposed imaging technique holds potential for accurate tumor detection in breast cancer screening.

The primary goal of our proposed research project is to develop a cutting-edge 3D system for diagnosing malignancies. It has 12 antennas arranged in a circle around the phantom. The imaging system's antenna array serves as a sensor for the human body, sending and receiving microwave signals. Eventually, a 3D system with 12 antennas positioned in a circle around the phantom to cover all breast locations is suggested. The targets are precisely located and clearly recognized, effectively demonstrating the effectiveness of the built system as a potential breast cancer detection tool.

3 Antenna design

The objective of this study was to develop a UWB antenna for breast cancer detection. The proposed antenna is a simple rectangular patch antenna with a 50 Ω power supply. The antenna was designed and fabricated on an FR-4 substrate ($\varepsilon r = 4.3$ and $\sigma = 0.02$). We chose the FR4 substrate because it is commercially available, inexpensive, and easy to manufacture. The proposed antenna covers the entire FCC frequency band from 3.1 to 11.6 GHz dedicated to medical applications. It is illustrated in Fig. 1 [31, 32].

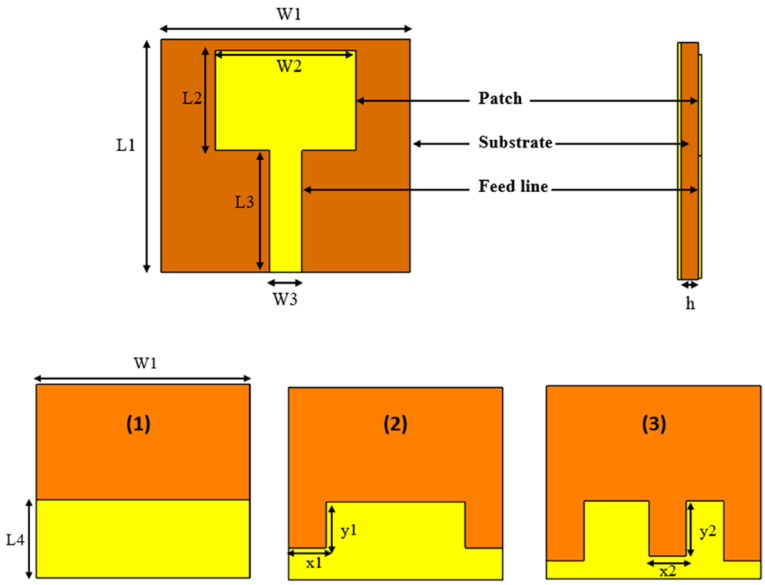

Fig. 1 The proposed antenna, step 1,2, and 3

Table 1 Antenna dimensions

Parameters	Dimensions (mm)	Parameters	Dimensions (mm)
W1	23	L4	8.5
L1	21	h	1.58
W2	13	×1	4
L2	9	y1	6.5
W3	3	×2	4
L3	11	y2	6

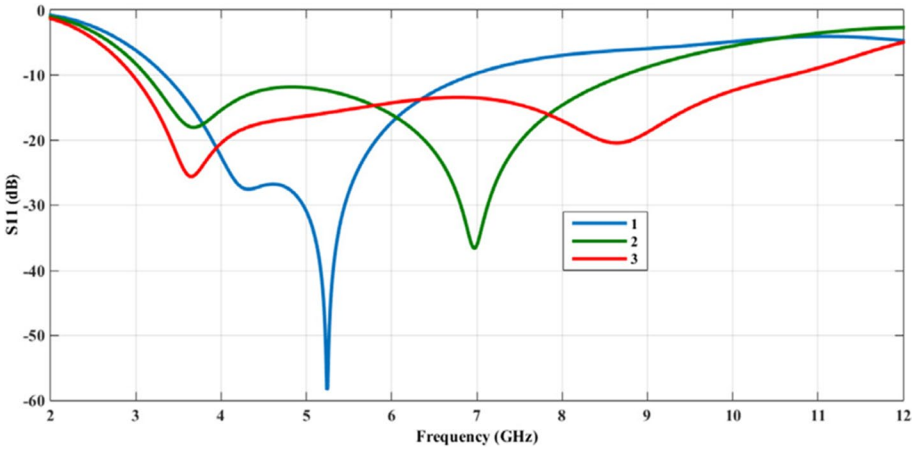

Fig. 2 Simulated reflection coefficient of the antenna for different steps

The evolution of antenna design includes three main steps illustrated in Fig. 1.

In step 1, there is a simple rectangular patch designed with L1 length and W1 width.

In step 2, the ground plane is the main factor affecting the performance of UWB antennas. The ground plane size can both expand the antenna's bandwidth from a few MHz to GHz and affect its performance. The partial ground plane technique allowed us to expand the bandwidth and note the impedance stability and therefore the quality of adaptation improves.

In Step 3: the slot technique one of the most widespread techniques that acts on the geometry of the antenna to reduce its size and improve its characteristics. The results show that slots and truncations lead to an increase in bandwidth from 2.96 to 10.68 GHz. Finally, after this series of optimization of the different parameters that allow us to expand the bandwidth, we came up with the final geometry of our antenna that meets our needs and objectives. The optimal dimensions of our antenna are shown in Table 1.

Figure 2 shows that the technique of inserting the slots into the ground plane with appropriate dimensions has a remarkable effect on the performance of our antenna. It allowed us to obtain two resonance frequencies at 3.65 GHz and 8.64 GHz with a maximum level of the reflection coefficient at -25.60 dB and -20.45 dB respectively, and a wide bandwidth ranging from 2.96 GHz to 10.68 GHz at -10 dB. Figure 3 shows the

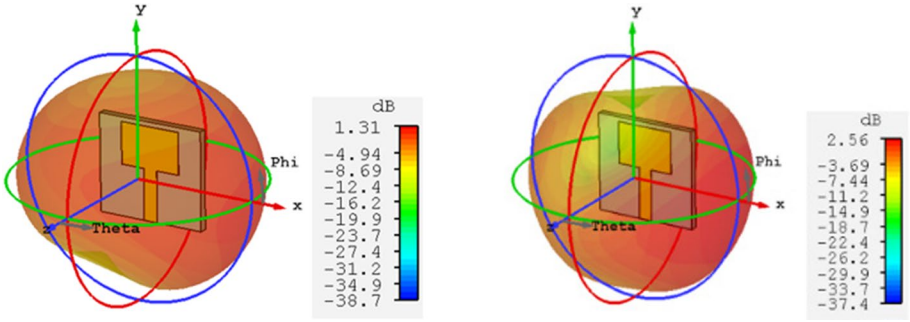

Fig. 3 3D radiation patterns at different frequencies of the proposed antenna, 3.65 and 8.64 GHz

radiation patterns in 3D. They make it possible to visualize the evolution of the radiation for the two frequencies 3.65 GHz and 8.64 GHz.

Figure 4 shows how the gain changes with frequency. Throughout the spectrum of frequencies that are relevant to us, from 3.1 GHz to 11.6 GHz, we have achieved good results suitable for FCC applications. At 11.6 GHz, we hit a peak gain of around 4.1 dB, showing strong performance in that specific range.

Figure 5 shows the realization of the proposed antenna. It offers the benefits of being compact in size, straightforward to manufacture, providing a wider bandwidth, featuring a simple design, delivering strong performance, and fully meeting FCC requirements and specifications.

Figure 6 shows the comparison of the coefficient of reflection obtained by simulation and realization. A good agreement is observed between the simulations and the measurements, or even a good impedance adaptation that varies around 50 Ω over the entire bandwidth.

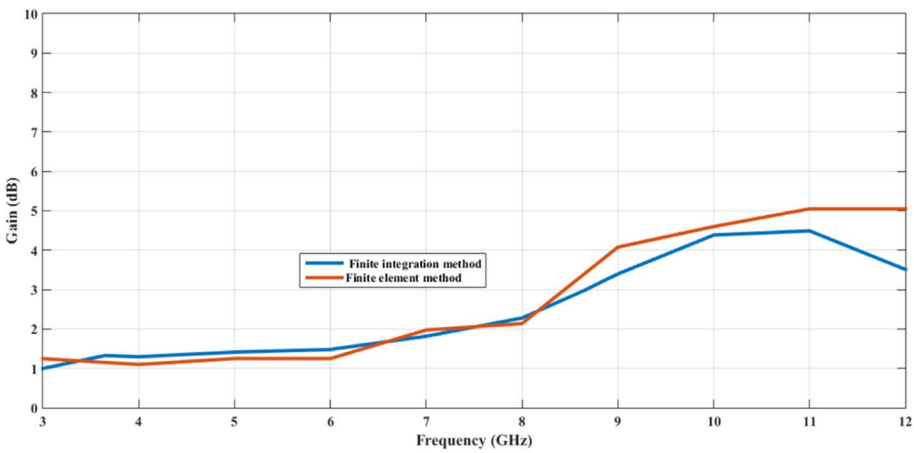

Fig. 4 Gain of the proposed antenna

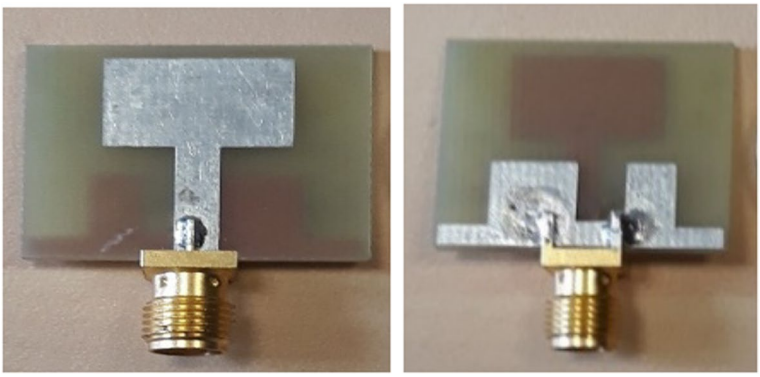

Fig. 5 Realized antenna: (1) top side, (2) bottom side

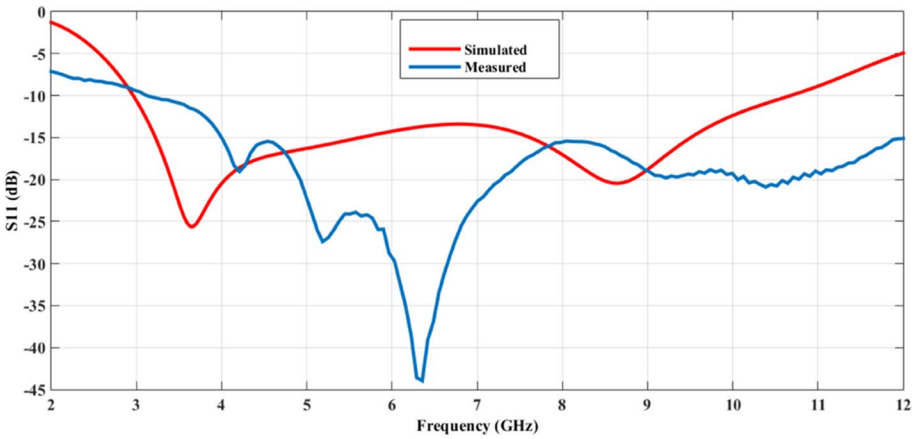

Fig. 6 Compared return loss of the UWB proposed antenna

4 Breast model

The breast model developed in this paper is a heterogeneous model of several biological tissues in half-sphere form, as shown in Figs. 7 and 8. Each tissue is characterized by frequency-dependent dielectric properties. As well as, to analyze the interaction between microwaves and tissues, the dielectric properties of the breast phantom must be close to that of the real. In a recent investigation on dielectric properties conducted by C. Gabriel [33, 34], it was demonstrated that a female breast can be accurately modeled using distinct tissues defined by highly specific parameters, such as: conductivity and permittivity matching the properties documented in real tissues. This method was developed and based on Eq. 1 [33, 34]. Table 2 illustrates the dielectric properties values for the layers of the breast (Skin and Fat) and tumor phantom.

$$\varepsilon_{\omega} = \varepsilon_{\infty} + \frac{\varepsilon_s - \varepsilon_{\infty}}{1 + (j\omega\tau)^{1-\alpha}} + \frac{\sigma}{j\omega\varepsilon_0} \tag{1}$$

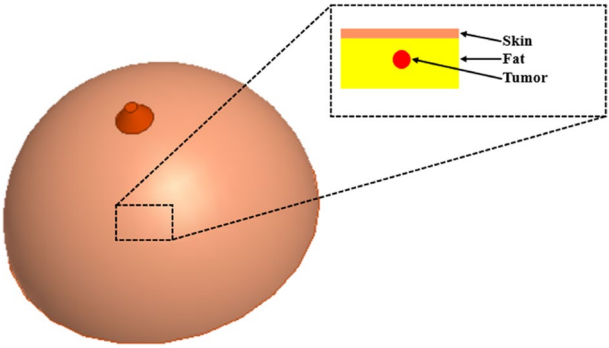

Fig. 7 Breast Model

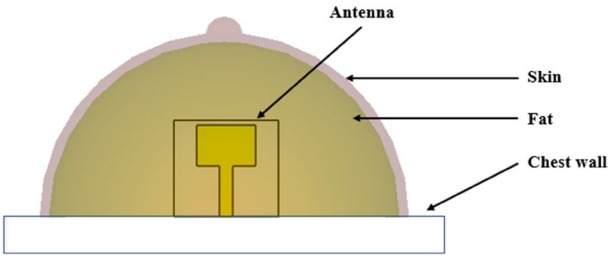

Fig. 8 Breast Model with proposed antenna

Table 2 The dielectric properties of breast tissue at 3 GHz

Tissues	Relative permittivity (ε_r)	Conductivity σ (S/m)	Tissue density ρ (Kg/m^3)
Skin	37.5	1.74	1109
Fat	5.04	0.179	911
Tumor	50	4	84

Where,

ω represents the angular frequency, α corresponds to the exponent parameter, ε_∞ signifies the relative permittivity, τ stands for the time constant, ε_s denotes the static relative permittivity, σ represents the static conductivity, and ε_0 is the permittivity of free space.

An antenna and its environment are usually interconnected, each affects the other, so there will always be a risk that the characteristics of the antenna will be disturbed by external factors. Biological tissues have a profound impact on the electromagnetic properties of an antenna, making them a critical factor to account for, consequently a study will be made to determine the effects of these biological tissues on the characteristics of our antenna.

For a portable antenna, the effect of the breast must also be considered, as these devices often must operate close to a human body. Figure 9 shows the reflection coefficient of the proposed antenna which is placed away from the breast at a distance d. When the antenna is positioned 0.5 mm from the breast, the − 10 dB bandwidth of the antenna widens, and the resonance point moves to the left. Although the antenna resonance frequencies change significantly in different scenarios, the proposed antenna UWB characteristic ensures S11-consistent performance of values below − 10 dB in the operational range for biomedical applications.

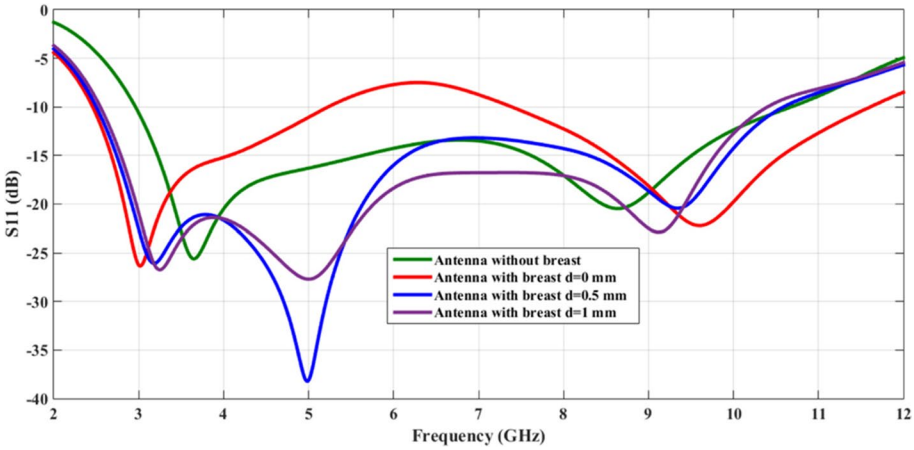

Fig. 9 Reflection coefficient of the proposed antenna with and without breast

5 Breast molds

Two molds have been designed and fabricated to make the layers of a breast so that their dielectric properties are close to those of a real breast, as shown in Figs. 10 and 11.

The first mold is made to make the layer of fat, is a half-sphere of radius 98 mm.

The second mold is made to make the layer of the skin and it includes two molds with a difference of 2 mm in their spokes. The smallest has a radius of 98 mm and the largest 100 mm. Both molds are designed to have a gap of 2 mm to properly create the skin layer.

In this part, we will make two mammary Phantoms, one with tumor and the other without tumor. The choice of chemicals used in the mixture of Phantoms was proposed by M. Lazebnik [35]. They have dielectric properties identical to those of real breast tissue, easy to find and mostly stable for long periods. Figure 12 illustrates the realization of three layers of breast (Skin, Fat, and Tumor).

Fig. 10 Mold design

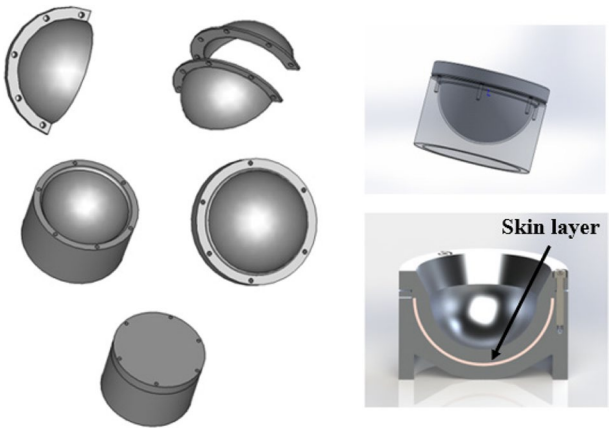

Fig. 11 Realization of molds

Fig. 12 Realization of phantom

6 System composition

Different configurations of 3D circular antenna array were initially simulated. After optimization, 12 elements are selected. The main idea is to incorporate as many elements as possible as long as the coupling between the elements is within the acceptable range. This will help us increase the accuracy and resolution of the constructed image. Having a

greater number of elements simplifies the data collection process at various angular positions, eliminating the need for unnecessary system rotations.

The 3D system designed is a half-sphere with a radius of 210 mm. The material chosen is an ABS plastic (acrylonitrile butadiene styrene) which has a smooth surface and easy to fabricate. Figure 13 illustrates the 3D design of each component of our system, it is composed of 5 pieces which can be gathered to create our final system. All components of our system are manufactured using a 3D printer. Figure 14 shows the system fabricated for breast cancer detection.

7 Results and discussion

The architecture of our breast cancer detection system is shown in Figs. 15 and 16. The concept of our system is to place our antenna on the breast and collect data for several positions of the antenna in order to cover the entire breast. After the end of the data collection an appropriate algorithm implemented in the laptop will process the signals to generate a 2D image of the breast with the exact position of the tumor, as reflected in Fig. 15. The total size of the detector is $21 \times 21 \times 12$ cm^3. This system consists of an antenna array, antenna carrier, power cables, vector network analyzer and PC. The antenna array is connected to the vector network analyzer via RF power cables, and it is connected to a laptop by a USB cable to exchange data and send commands. Patients are asked to lie on a bed and the system is placed below the breast, as shown in Figs. 16 and 17.

The transmitting antenna is selected first, and then the receiving antennas are selected in turn from 1 to 12. Then, the transmitting antenna is modified, and the receiving antennas are selected successively, as shown in Fig. 18. During the operation, we prevented

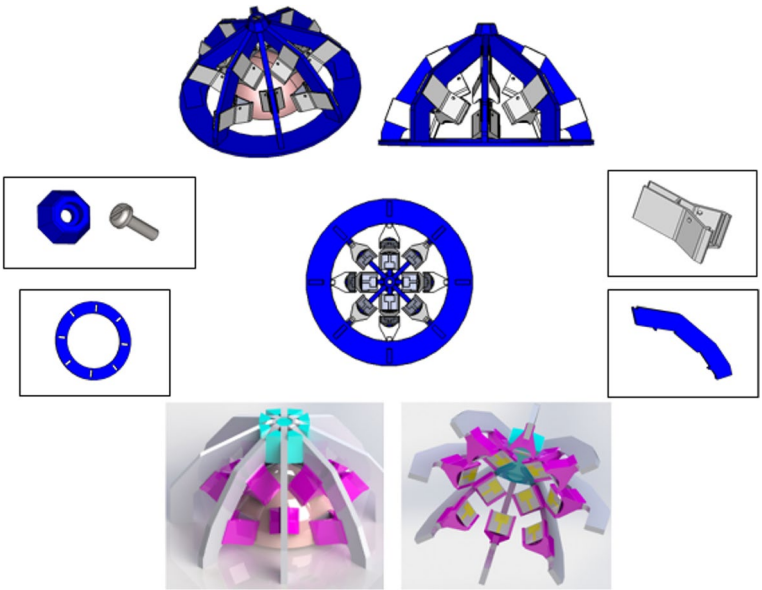

Fig. 13 3D system design

Fig. 14 Realization of the 3D system

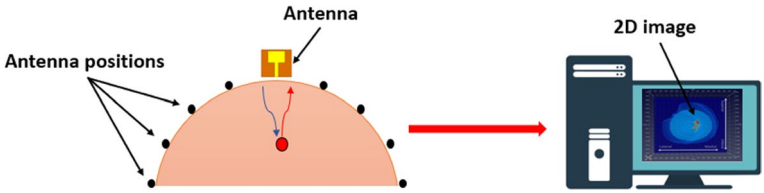

Fig. 15 Process of our proposed system

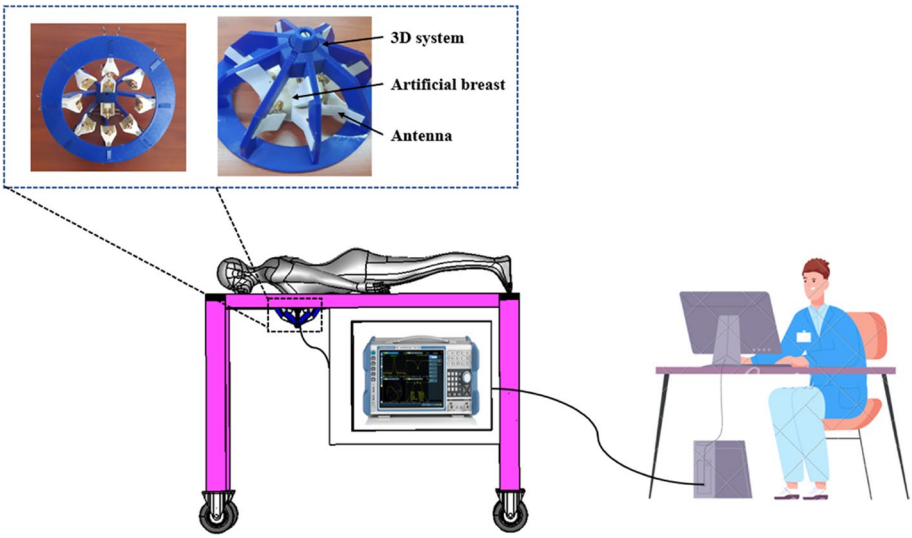

Fig. 16 Proposed configuration of our system

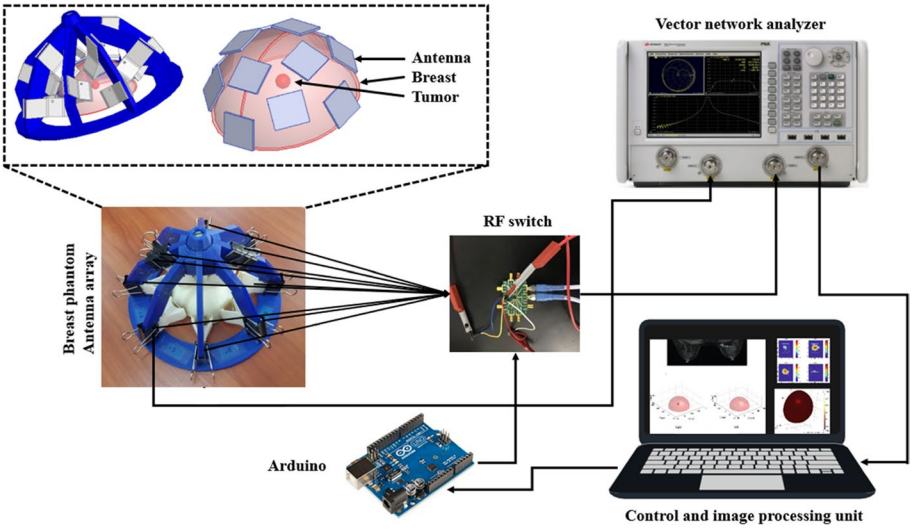

Fig. 17 Various elements of the suggested breast imaging setup

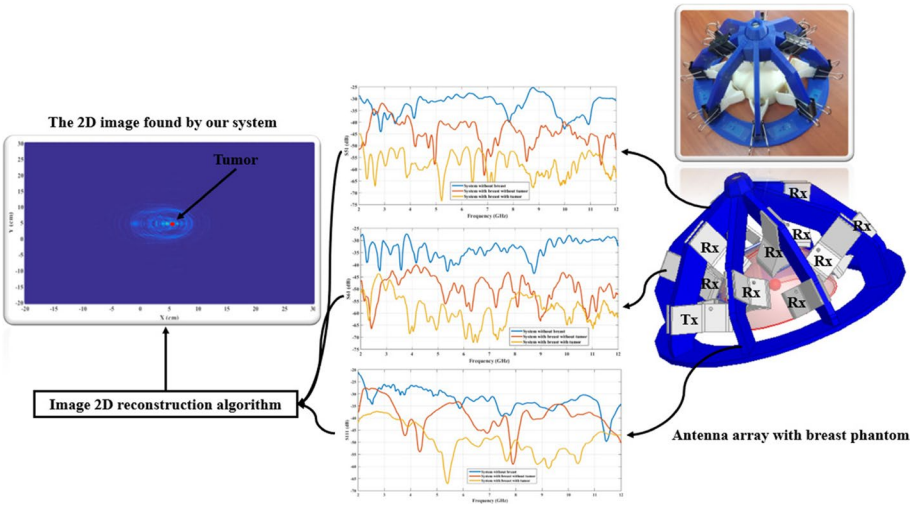

Fig. 18 Breast cancer detection system and 2D image reconstruction

the coaxial cables from being twisted, which can lead to a large variation in the signals received.

At the start of each testing procedure, patients are gently instructed to recline upon a bed, and our specialized system is methodically placed beneath the breast, aligning with the instructions illustrated in Fig. 16. This precise positioning is integral to ensuring the system's optimal interaction with the breast tissue during the examination. It serves as a crucial foundation for acquiring dependable and precise data throughout the entirety of the testing, facilitating a thorough and accurate evaluation of the breast's condition.

In our testing, we didn't employ a real woman, instead, we utilized crafted artificial breasts that closely mimic the physical characteristics of a real woman's breasts. These artificial breasts have been painstakingly designed and manufactured to replicate the size, shape, density, and tissue composition of real breast tissue. This meticulous attention to detail was crucial to ensure that our testing conditions closely approximate the real-life scenarios encountered in breast examinations.

The testing procedure commences with the deliberate selection of antenna 1 for transmission, while antennas 2 through 12 are strategically chosen for reception purposes. A critical aspect of this process is the careful preservation of all data acquired throughout the test until its culmination. During the test's execution, the transmitting antenna 1 is chosen, while the receiving antennas are chosen one after another. This meticulous sequence of operations is central to the objective of accumulating an extensive dataset that captures the transmission coefficient for each individual antenna and its corresponding position in space.

The transmission coefficient itself holds a pivotal role in this experimental framework. It serves as a quantitative measure that precisely quantifies the effectiveness with which the transmitted signal is received and detected by the array of antennas. This metric is essential for understanding how the electromagnetic waves emitted by the transmitting antenna interact with the surrounding environment and the receiving antennas, shedding light on signal propagation, absorption, and reflection characteristics.

Upon the successful conclusion of the test, a comprehensive set of results materializes in the form of graphical representations. These representations manifest as three distinct and vividly color-coded curves. The blue curves, representing measurements obtained when the system operates in an antenna-only configuration without any breast tissue, establish a crucial baseline for the system's performance. These measurements help understand how the antennas and the environment interact without the presence of biological tissue.

The red curves provide a contrasting perspective by encapsulating measurements obtained from a breast that is free of tumors. This "tumor-free" breast serves as a reference point for understanding how healthy breast tissue impacts the transmission coefficient.

Lastly, the orange curves depict measurements acquired when the system encounters a breast with tumors. These curves offer crucial insights into the influence of anomalies within the breast tissue on the transmission coefficient. We can use these data to discern the effects of tumors, their size, and their location on the behavior of electromagnetic signals within the breast.

In sum, this intricate and systematic testing approach enables scientists and medical professionals to gain a comprehensive understanding of how electromagnetic signals interact with breast tissue, providing essential insights for breast health assessment and early detection of abnormalities.

Figures 19 and 20 illustrate the transmission coefficient between antennas 1 and 2 for three different environments across the entire FCC band from 3.1 to 10.6 GHz. In this case, it can be observed that the blue curve of the environment without a breast is around -25 dB, which is a tolerable value indicating that the antenna array meets all the desired standards. This means that all antennas can operate without affecting the characteristics of the nearby antennas, and we can consider this curve as a reference compared to the other cases 2 and 3. For case 2, represented by the red curve, we notice that the transmission coefficient has decreased to around -30 dB. The presence of the breast has caused this decrease, indicating that the antennas received less energy compared to the first case. The system detected the added layers (skin, fat …). In case 3, represented by the orange curve, we observe that the transmission coefficient has decreased to around -40 dB. The presence of the tumor

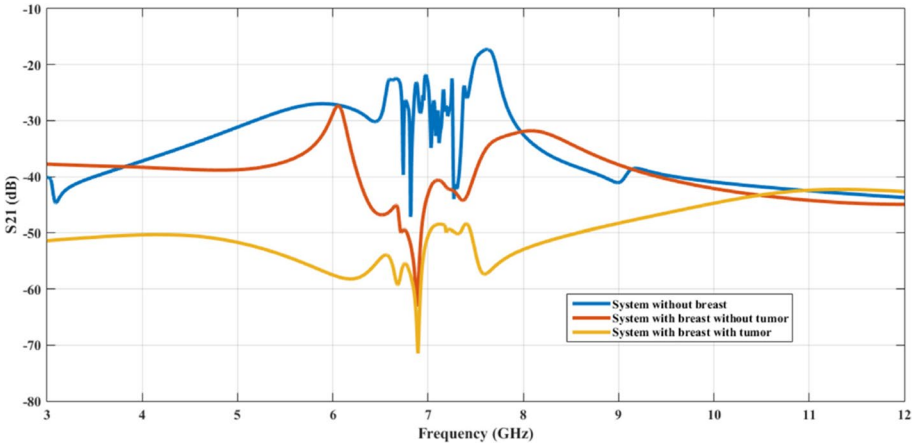

Fig. 19 Transmission coefficient S2,1 simulated

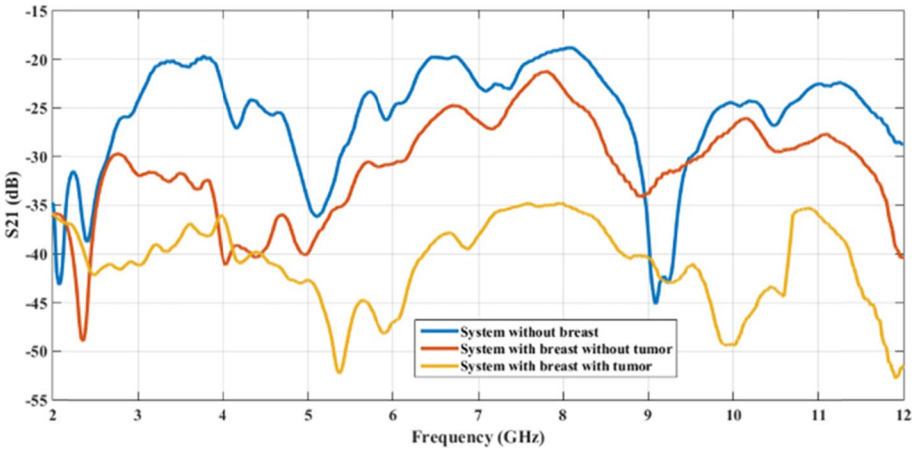

Fig. 20 Transmission coefficient S2,1 measured

has caused this decrease due to its high dielectric properties compared to the other layers of the skin and fat. Therefore, it is clear that the presence of a tumor or other layers results in a decrease in coupling between antennas 1 and 2, characterized by a weakened signal compared to the first case. This decreased coupling is a key indicator that our system can effectively identify any added layers, especially the presence of a tumor in breast tissue.

Figures 21 and 22 provide a visual representation of the transmission coefficient between antennas 1 and 3 across three different scenarios. In this context, the blue curve corresponding to the environment without a breast exhibits a value of around -35 dB. This value is well within acceptable limits, indicating that our antenna array complies with all the required standards. It demonstrates that all antennas can function without affecting the performance of nearby antennas, and we can regard this curve as a baseline when comparing it to cases 2 and 3. In the case represented by the red curve (case 2), we observe a drop in the transmission coefficient to approximately -45 dB. This decline can be attributed to the presence of the breast, which means that the antennas received less energy compared to

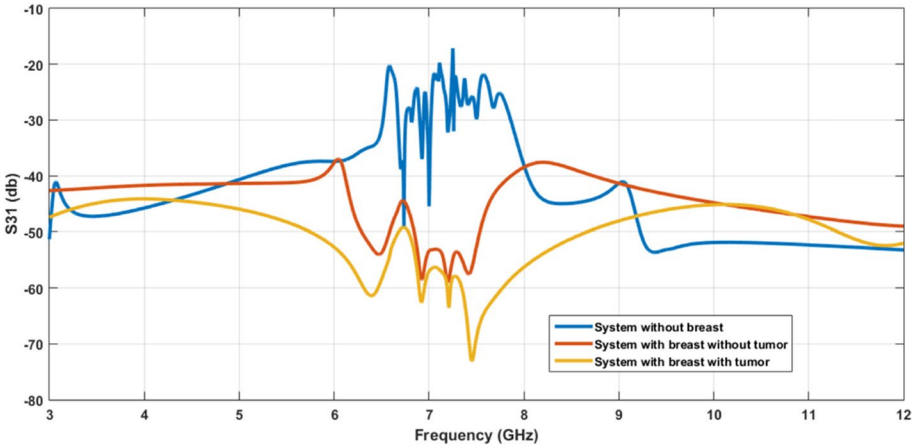

Fig. 21 Transmission coefficient S3,1 simulated

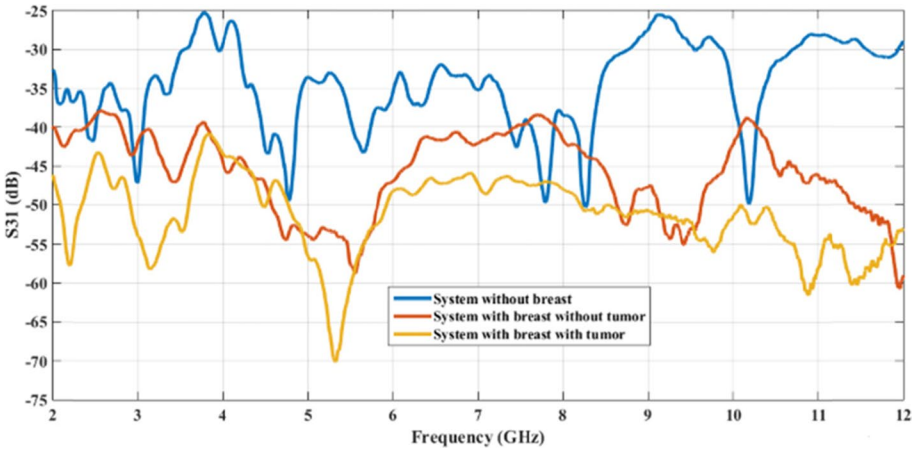

Fig. 22 Transmission coefficient S3,1 measured

the first scenario. The system effectively identified the additional layers, such as skin and fat. Moving on to case 3, denoted by the orange curve, we notice a further decrease in the transmission coefficient to roughly -50 dB. The presence of a tumor caused this drop due to its significantly different dielectric properties compared to the other layers of skin and fat. Therefore, it is evident that the introduction of a tumor results in a reduction in the coupling between antennas 1 and 3, resulting in a weaker signal compared to the first scenario.

Figures 23 and 24 serve as visual aids to depict the transmission coefficient between antennas 1 and 4 in three distinct settings. In this context, the blue curve representing the environment without a breast around -35 dB. This value is well within acceptable limits, indicating that our antenna array adheres to all the requisite standards. It implies that all antennas can operate without interfering with the performance of neighboring antennas, and we can use this curve as a benchmark when comparing it to cases 2 and 3. In the case depicted by the red curve (case 2), we observe a decrease in the transmission coefficient to approximately -45 dB. This reduction can be attributed to the presence of the breast, which

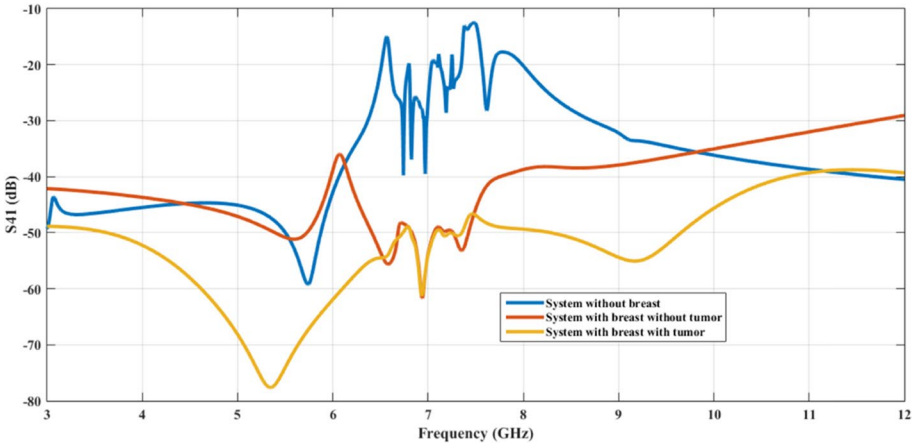

Fig. 23 Transmission coefficient S4,1 simulated

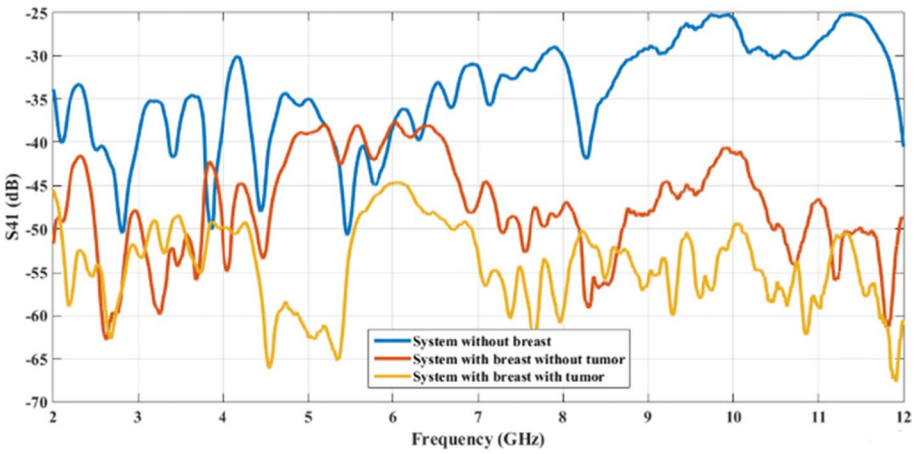

Fig. 24 Transmission coefficient S4,1 measured

means that the antennas received less energy compared to the first scenario. The system adeptly detected the additional layers. As for case 3, illustrated by the orange curve, we note a further decrease in the transmission coefficient to roughly -50 dB. The presence of a tumor caused this decline, owing to its markedly different dielectric properties compared to the other layers of skin and fat. Therefore, it becomes evident that the introduction of a tumor or other layers results in a diminished coupling between antennas 1 and 4, leading to a weaker signal compared to the first scenario.

Figures 25 and 26 provide graphical representations of the transmission coefficient between antennas 1 and 5 under three different environmental conditions. In this context, the blue curve, which corresponds to the environment without a breast, maintains a level of approximately -30 dB. This value falls within an acceptable range, indicating that our antenna array complies with all the relevant standards. It demonstrates that all antennas can operate without interfering with the performance of nearby antennas, and we can

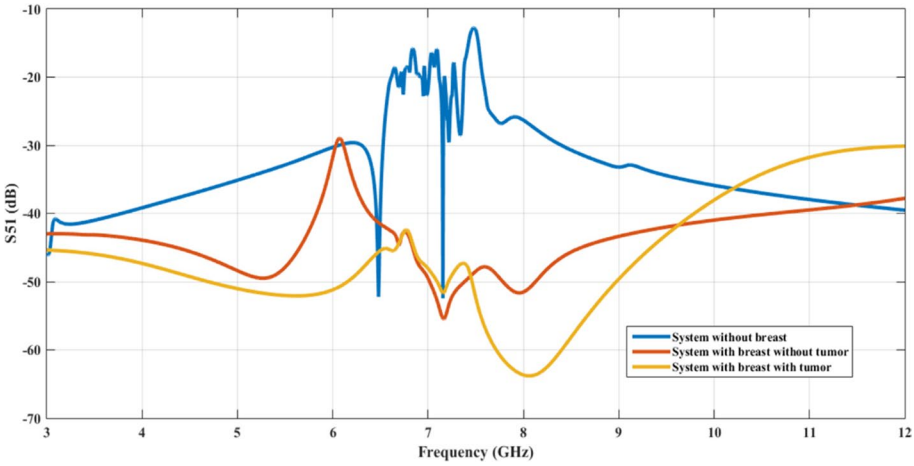

Fig. 25 Transmission coefficient S5,1 simulated

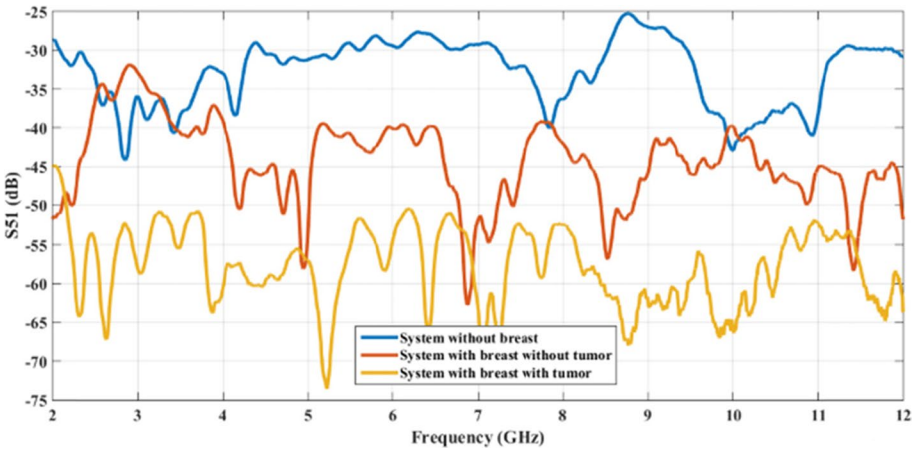

Fig. 26 Transmission coefficient S5,1 measured

use this curve as a reference point when comparing it to cases 2 and 3. In the case represented by the red curve (case 2), we observe a reduction in the transmission coefficient to around -45 dB. This decrease can be attributed to the presence of the breast, signifying that the antennas received less energy compared to the first scenario. The system effectively detected the additional layers. In the case of case 3, depicted by the orange curve, we notice a further decline in the transmission coefficient to approximately -55 dB. This decline is caused by the presence of a tumor, which has significantly different dielectric properties compared to the other layers of skin and fat. Consequently, it is evident that the introduction of a tumor or other layers results in a decreased coupling between antennas 1 and 5, resulting in a weaker signal compared to the first scenario.

Figures 27 and 28 provide graphical representations of the transmission coefficient between antennas 1 and 6 in three distinct scenarios. In the case of the blue curve, which represents the environment without a breast, the coefficient hovers around -35 dB. This

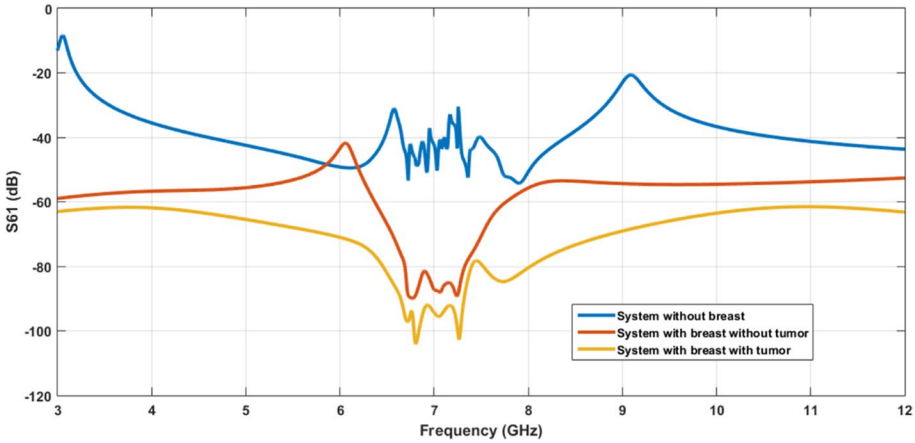

Fig. 27 Transmission coefficient S6,1 simulated

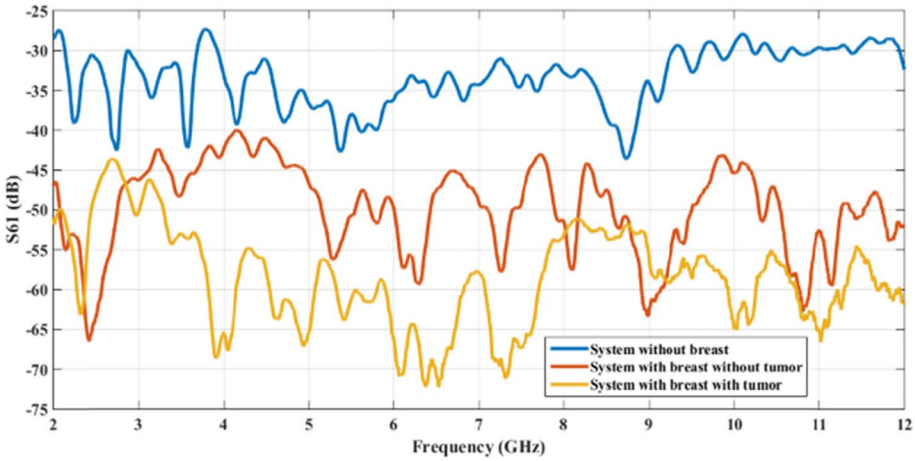

Fig. 28 Transmission coefficient S6,1 measured

value is well within acceptable limits, signifying that our antenna array conforms to all the necessary standards. It indicates that all antennas can function without causing interference to nearby antennas. This blue curve serves as a baseline when we compare it to cases 2 and 3. In the case depicted by the red curve (case 2), we observe a reduction in the transmission coefficient to approximately -50 dB. The presence of the breast is responsible for this decrease, indicating that the antennas received less energy compared to the first scenario. For case 3, illustrated by the orange curve, we note a further decline in the transmission coefficient to roughly -60 dB. This decline is attributed to the presence of a tumor, which exhibits significantly different dielectric properties. Consequently, it becomes evident that the introduction of a tumor or other layers results in reduced coupling between antennas 1 and 6, leading to a weaker signal compared to the first scenario.

Figures 29 and 30 visually represent the transmission coefficient between antennas 1 and 7 under three distinct environmental conditions. In the context of the blue curve,

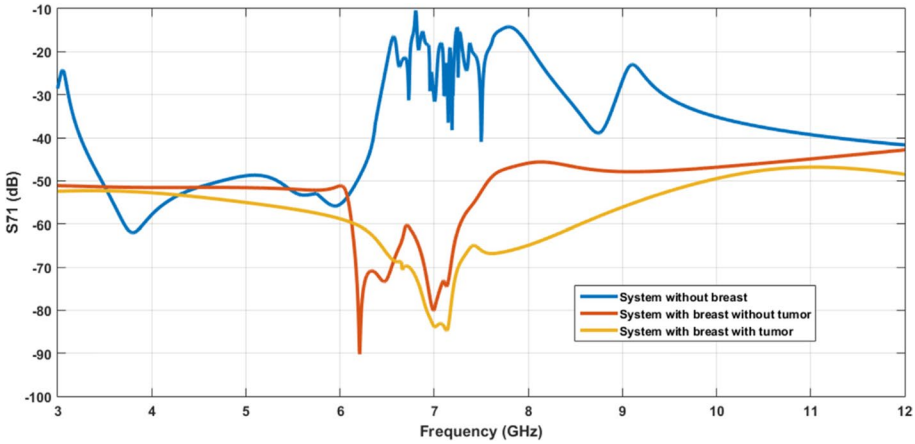

Fig. 29 Transmission coefficient S7,1 simulated

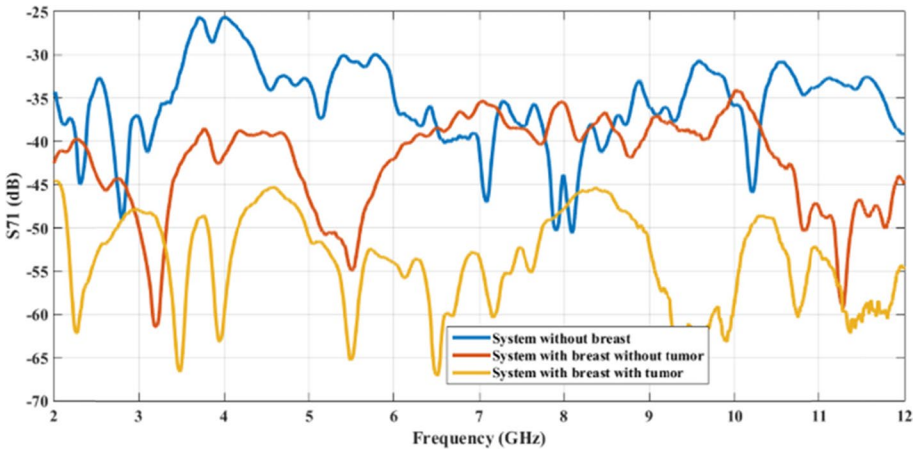

Fig. 30 Transmission coefficient S7,1 measured

which signifies the environment without a breast, the coefficient remains approximately at -35 dB. This value is comfortably within acceptable parameters, affirming that our antenna array aligns with all necessary standards. It underscores that all antennas can operate without causing interference to their nearby counterparts, and we use this curve as a reference point when comparing it to cases 2 and 3. In the case portrayed by the red curve (case 2), we observe a dip in the transmission coefficient to roughly -40 dB. This reduction can be attributed to the presence of the breast, indicating that the antennas received less energy compared to the first scenario. The system effectively discerned the additional layers, encompassing the skin and fat. As for case 3, represented by the orange curve, we discern a further descent in the transmission coefficient to around -50 dB. This drop is a result of the presence of a tumor. Thus, it becomes evident that the introduction of a tumor or other layers leads to a decreased coupling between antennas 1 and 7, culminating in a weaker signal relative to the first scenario.

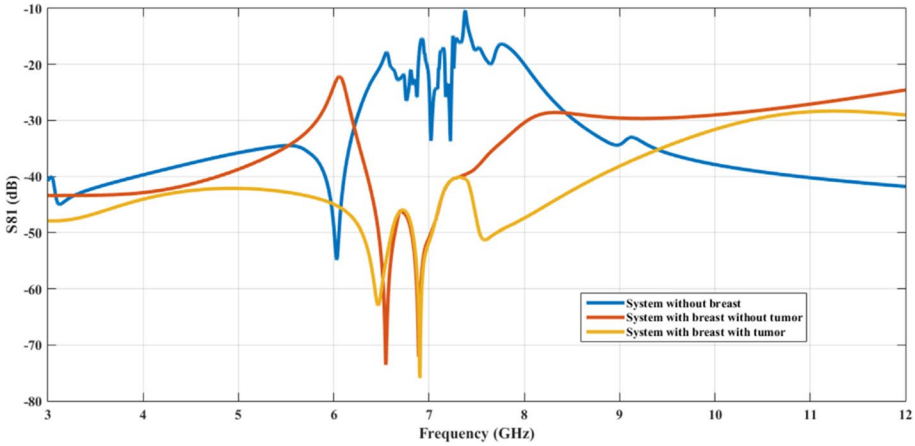

Fig. 31 Transmission coefficient S8,1 simulated

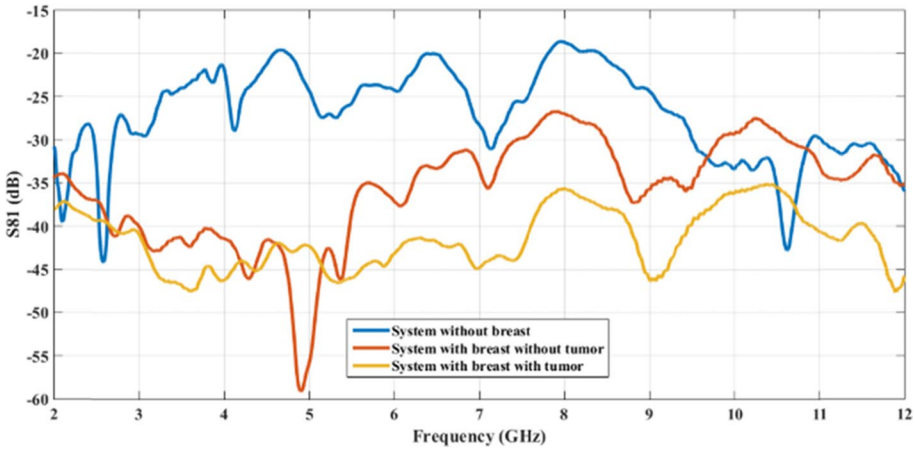

Fig. 32 Transmission coefficient S8,1 measured

Figures 31 and 32 visually depict the transmission coefficient between antennas 1 and 8 under three different environmental settings. In the case of the blue curve, representing the environment without a breast, the coefficient remains at approximately -25 dB. This value falls well within acceptable limits, affirming that our antenna array complies with all relevant standards. It signifies that all antennas can operate without causing interference to neighboring antennas, serving as a benchmark for comparison with cases 2 and 3. In the case portrayed by the red curve (case 2), we observe a reduction in the transmission coefficient to around -35 dB. The presence of the breast is responsible for this decrease, indicating that the antennas received less energy compared to the first scenario. For case 3, represented by the orange curve, we notice a further decline in the transmission coefficient to approximately -45 dB. This drop is attributed to the presence of a tumor, which possesses significantly different dielectric properties compared to the other layers of skin and fat.

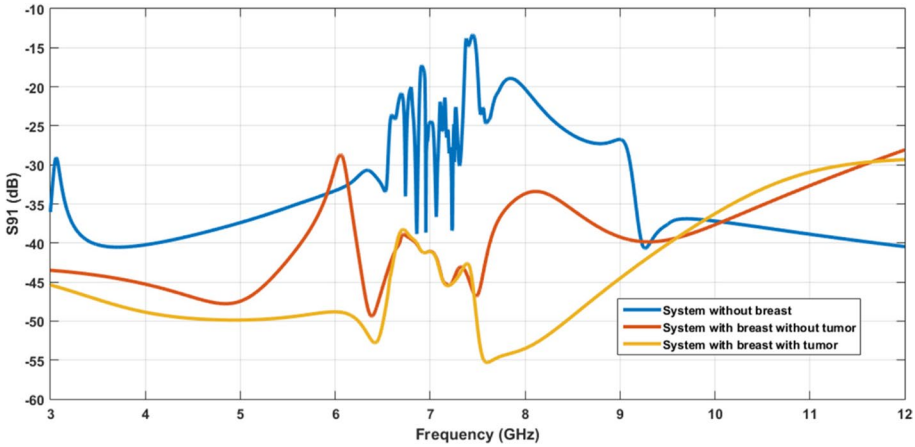

Fig. 33 Transmission coefficient S9,1 simulated

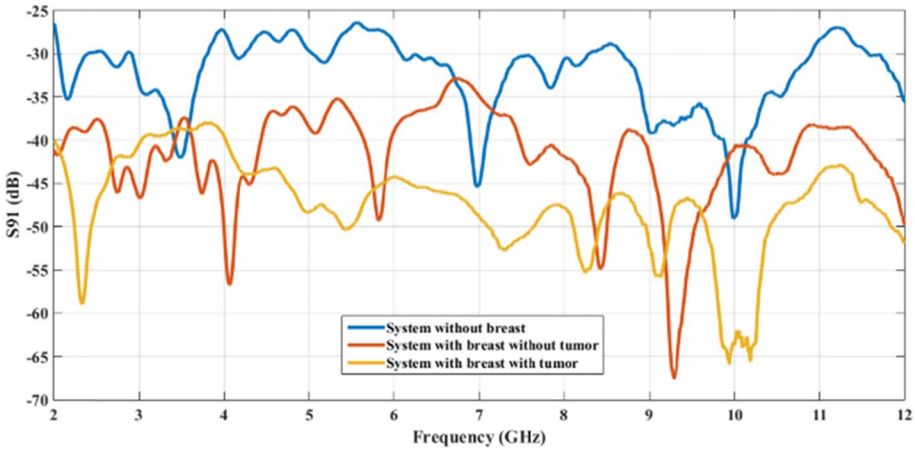

Fig. 34 Transmission coefficient S9,1 measured

Figures 33 and 34 provide graphical representations of the transmission coefficient between antennas 1 and 9 in three distinct scenarios. In the case of the blue curve, which corresponds to the environment without a breast, the coefficient remains stable at around -30 dB. This value comfortably aligns with acceptable standards, demonstrating that our antenna array adheres to the required norms. It underscores that all antennas can function without interfering with the performance of neighboring antennas, serving as a reference point when comparing it to cases 2 and 3. In the case depicted by the red curve (case 2), we observe a reduction in the transmission coefficient to approximately -40 dB. This decline can be attributed to the presence of the breast, indicating that the antennas received less energy compared to the first scenario. As for case 3, represented by the orange curve, we observe a further decrease in the transmission coefficient to roughly -45 dB. This decrease is linked to the presence of a tumor. Therefore, it becomes evident

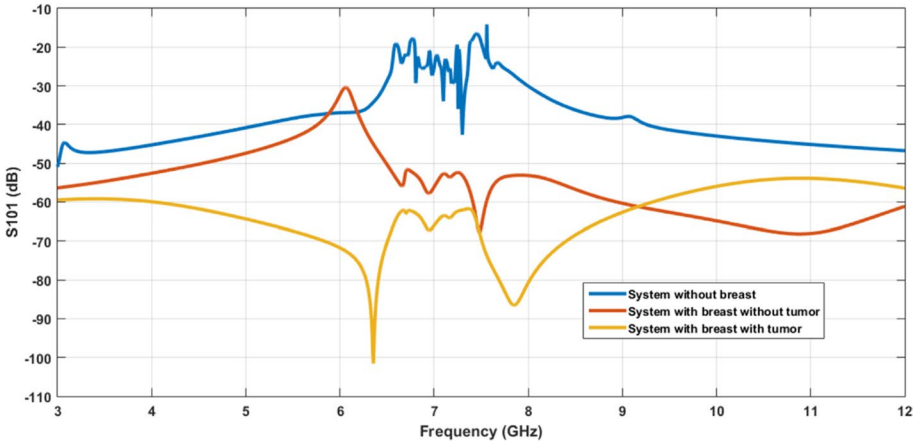

Fig. 35 Transmission coefficient S10,1 simulated

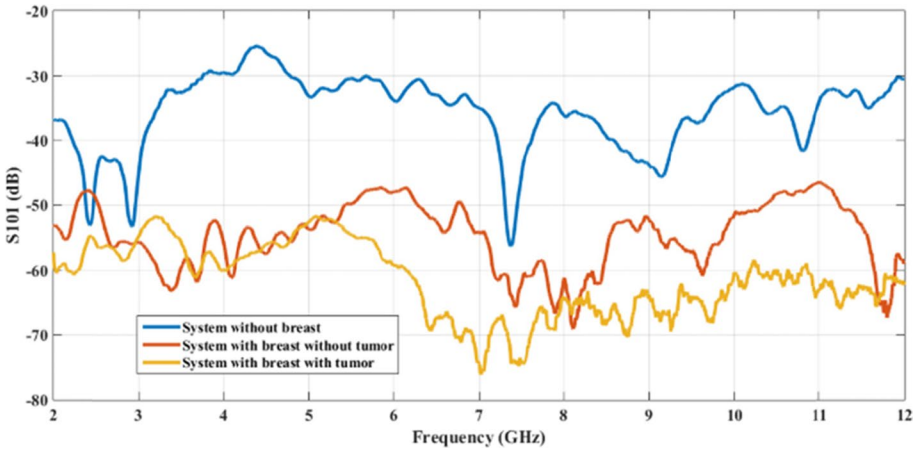

Fig. 36 Transmission coefficient S10,1 measured

that the introduction of a tumor or other layers leads to reduced coupling between antennas 1 and 9, resulting in a weaker signal compared to the first scenario.

Figures 35 and 36 visually demonstrate the transmission coefficient between antennas 1 and 10 across three distinct environmental conditions. In the case of the blue curve, representing the environment without a breast, the coefficient remains relatively stable at around -30 dB. This value comfortably meets acceptable standards, affirming the compliance of our antenna array with necessary criteria. It signifies that all antennas can operate without causing interference to neighboring antennas, serving as a benchmark for comparison with cases 2 and 3. In the scenario depicted by the red curve (case 2), we observe a drop in the transmission coefficient to approximately -50 dB. This reduction can be attributed to the presence of the breast, indicating that the antennas received less energy compared to the first scenario. The system adeptly identified the additional layers. As for case 3, illustrated by the orange curve, we note a further decline in the transmission coefficient to roughly -60 dB. This decline is linked to the presence of a tumor, which possesses markedly

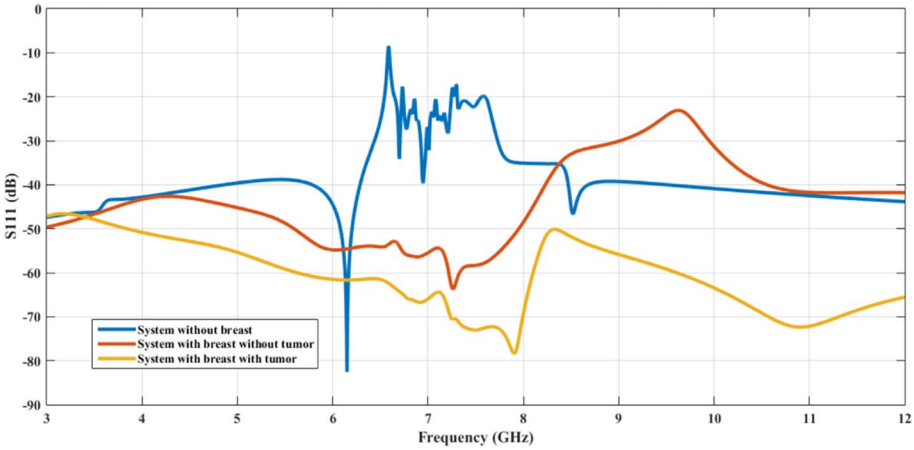

Fig. 37 Transmission coefficient S11,1 simulated

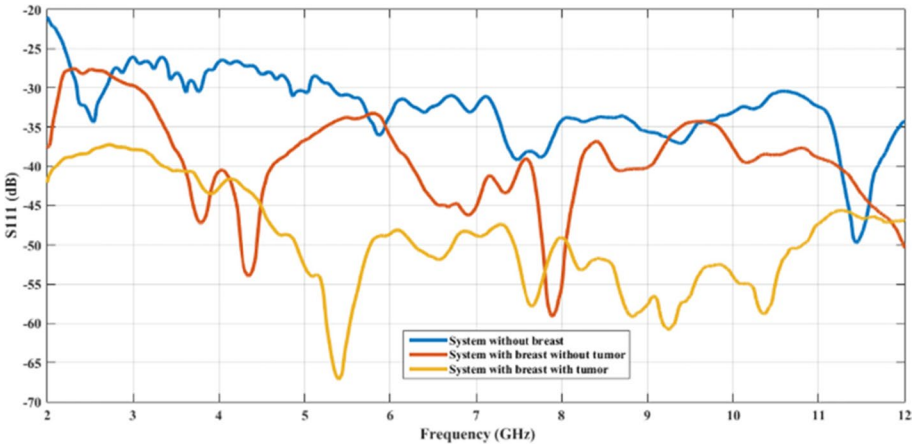

Fig. 38 Transmission coefficient S11,1 measured

different dielectric properties compared to the other layers of skin and fat. it is clear that the presence of the tumor has a significant effect on the level of coupling between antenna 1 and 10. This drop in coupling is due to the disturbance of the electromagnetic field caused by the tumor itself. The tumor can act as an obstacle that disrupts the propagation of electromagnetic waves and thus prevents coupling between the antennas.

In Figs. 37 and 38, we can visually observe the transmission coefficient between antennas 1 and 11 under three distinct environmental conditions. In the case of the blue curve, representing the environment without a breast, the coefficient consistently hovers around -30 dB. This value aligns well with the accepted standards, confirming that our antenna array complies with the necessary requirements. It underscores that all antennas can operate without causing interference to nearby antennas, serving as a point of reference for comparison with cases 2 and 3. In the scenario depicted by the red curve (case 2), we witness a reduction in the transmission coefficient to approximately -40 dB. This reduction is attributable to the presence of the breast, indicating that the antennas received less

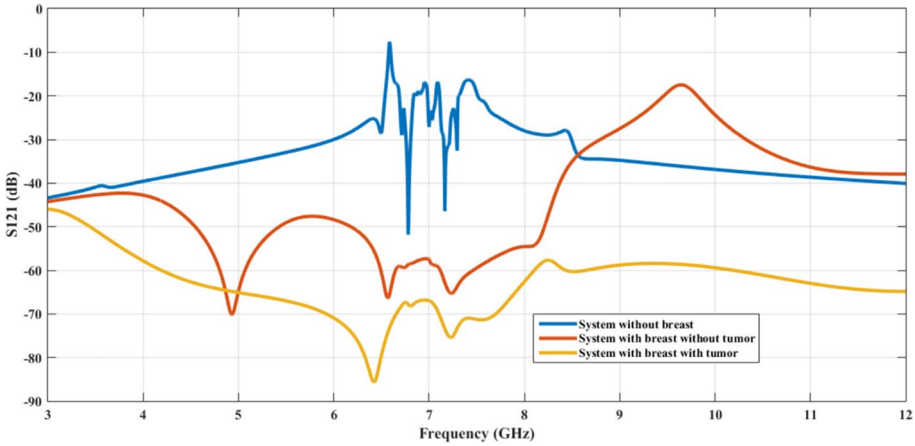

Fig. 39 Transmission coefficient S12,1 simulated

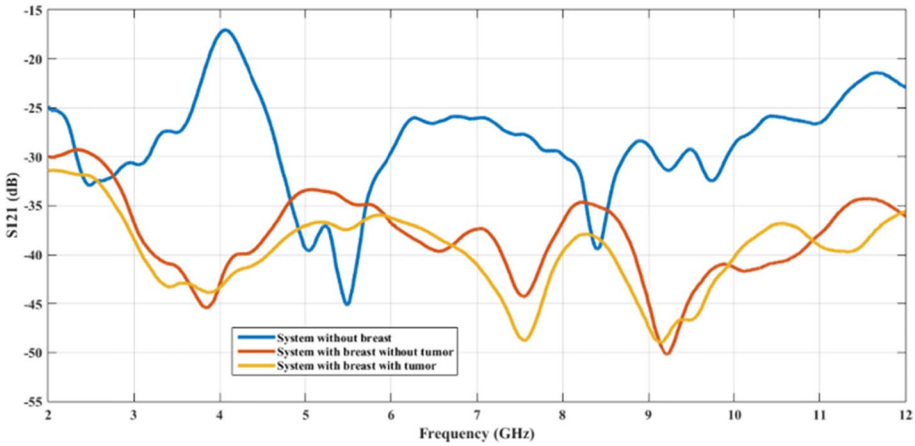

Fig. 40 Transmission coefficient S12,1 measured

energy compared to the first scenario. The system efficiently identified the additional layers, including the skin and fat. For case 3, illustrated by the orange curve, we observe a further decline in the transmission coefficient to approximately -50 dB. This decline can be attributed to the presence of a tumor, which possesses significantly different dielectric properties compared to the other layers of skin and fat.

Figures 39 and 40 offer a graphical representation of the transmission coefficient between antennas 1 and 12 in three distinct scenarios. In the case of the blue curve, which corresponds to the environment without a breast, the coefficient consistently hovers at around -30 dB. This value squarely fits within acceptable limits, affirming that our antenna array meets all requisite standards. It underscores that all antennas can function without causing interference to nearby antennas, and we use this curve as a benchmark for comparison with cases 2 and 3. In the scenario represented by the red curve (case 2), we observe a decrease in the transmission coefficient to approximately -35 dB. This decrease is attributed to the presence of the breast, indicating that the antennas received less energy

compared to the first scenario. The system adeptly identified the additional layers, including the skin and fat. As for case 3, illustrated by the orange curve, we note a further drop in the transmission coefficient to roughly -40 dB. This decline is linked to the presence of a tumor, which possesses markedly different dielectric properties compared to the other layers of skin and fat. The presence of the tumor has the potential to serve as a hindrance, disrupting the propagation of electromagnetic waves and consequently inhibiting the coupling between the antennas. This reduction in coupling serves as a pivotal indicator of our system's ability to effectively detect the presence of added layers, particularly tumors within breast tissue.

All of our simulations have consistently demonstrated the capability of our system to detect tumors by analyzing signals from three distinct environments: one devoid of a breast, one with a breast without tumor, and one with a breast containing a tumor. Notably, these simulations reveal that the presence of a tumor acts as an impediment within the breast tissue, resulting in a significantly reduced signal coupling compared to the other scenarios. This decreased coupling is a key indicator that our system can effectively identify the presence of a tumor in breast tissue by analyzing signals.

The measurement of the transmission coefficient is an important parameter for evaluating the performance of an antenna system. It helps to determine the amount of signal loss during transmission and reception. The test is crucial for evaluating the performance of an antenna system in breast imaging and detecting breast tumors.

The responses derived from transmitting antenna 1 exhibit significant variability contingent upon the surrounding environment. The highest recorded maximum transmission coefficient, approximately -50 dB, is observed in the absence of any phantom, highlighting the baseline performance of the system. In contrast, when a phantom mimicking a breast with a tumor is introduced, the maximum transmission coefficient markedly drops to around -75 dB across the FCC frequency band. This discrepancy in signal propagation can be attributed to the distinct dielectric properties of the tumor, which differ notably from those of normal breast tissue.

Furthermore, it is imperative to note the variations in antenna coupling across these scenarios. In the first case, a comparatively low coupling between the antennas is observed. In the second case, this coupling diminishes relative to the first case, indicating a shift in signal interaction. The third case witnesses a significant decrease in coupling compared to the previous two cases, a trend consistently observed across all figures. This cumulative evidence strongly suggests that the proposed antenna array system possesses the capability to detect additional layers, with a concomitant weakening of coupling as more layers are introduced. This underscores the system's suitability for breast imaging applications, as it effectively scrutinizes the transmitted and received signals, offering a promising avenue for the identification of breast tumors through comprehensive signal analysis.

Within the specified frequency range of interest (3.1 GHz to 11.6 GHz UWB according to FCC regulations), it has been observed that mutual coupling is influenced by several factors including distance, relative positions, and the surrounding environment, which encompasses various layers, as detailed in Table 3. To illustrate, Examining the mutual coupling between A1 and A2 in case 1. This coupling exhibits variations spanning from approximately -20 dB to -47 dB (with measured values falling in the range of -20 dB to -45 dB). In case 2, the mutual coupling ranges from about -30 dB to -62 dB (with measured values between -25 dB and -49 dB), while in case 3, it fluctuates between -50 dB and -71 dB (with measured values ranging from -35 dB to -52 dB).

The diminishing trend in mutual coupling across these cases can be attributed to the inclusion of multiple layers, specifically the presence of breast tissue with and without

Table 3 The various simulated and measured transmission coefficient values for the three studied cases

Transmission coefficient	Case 1: System without breast ≈ Min, Max values (dB)		Case 2: System with breast without tumor ≈ Min, Max values (dB)		Case 3: System with breast with tumor ≈ Min, Max values (dB)	
	Simulated	Measured	Simulated	Measured	Simulated	Measured
S21	≈ -20, -47	≈ -20, -45	≈—30, -62	≈ -25, -49	≈ -50, -71	≈ -35, -52
S31	≈ -20, -53	≈ -25, -49	≈ -40, -58	≈ -40, -60	≈ -45, -72	≈ -45, -70
S41	≈ -20, -59	≈ -25, -50	≈ -35, -61	≈ -40, -62	≈ -42, -78	≈ -45, -67
S51	≈ -20, -52	≈ -25, -44	≈ -30, -55	≈ -35, -60	≈ -42, -64	≈ -50, -72
S61	≈ -20, -47	≈ -30, -42	≈ -40, -84	≈ -40, -66	≈ -60, -102	≈ -45, -72
S71	≈ -20, -60	≈ -25, -50	≈ -48, -80	≈ -35, -62	≈ -51, -83	≈ -45, -67
S81	≈ -20, -51	≈ -20, -44	≈ -22, -73	≈ -26, -45	≈ -30, -77	≈ -35, -47
S91	≈ -20, -40	≈ -26, -45	≈ -30, -48	≈ -35, -55	≈ -33, -55	≈ -40, -65
S101	≈ -20, -48	≈ -30, -50	≈ -30, -70	≈ -50, -70	≈ -52, -100	≈ -53, -78
S111	≈ -20, -40	≈ -25, -40	≈ -22, -60	≈ -30, -56	≈ -49, -80	≈ -40, -65
S121	≈ -20, -40	≈ -20, -45	≈ -20, -70	≈ -30, -50	≈ -45, -85	≈ -35, -50

tumors. Remarkably, the measured values align closely with the outcomes of our simulations, indicating a strong correlation.

This pattern of mutual coupling behavior can similarly be observed across various positions and their corresponding transmission coefficients, denoted as Sn1 (where n = 2 to 12) as shown in Table 3.

In summary, the mutual coupling phenomenon in the specified frequency band demonstrates sensitivity to factors like distance, positions, and the complex environment, marked by the presence of different layers, such as breast tissues with and without tumors. This trend of decreasing mutual coupling is consistent across multiple scenarios and aligns well with our simulation results, a pattern that extends to other positions and their respective transmission coefficients. This highlights the system's ability to differentiate signals, ultimately facilitating the accurate detection of breast tumors by analyzing all signals from various positions.

8 Breast tumor detection using microwave confocal imaging method

Microwave Confocal Imaging (MCI) is an electromagnetic approach employed to identify the presence and precise location of breast tumors. It has been chosen for use in UWB microwave imaging due to its established reliability, straightforward implementation, robustness, and rapid computational capabilities [20, 36, 37]. MCI relies on a series of underlying assumptions concerning the dielectric properties of both malignant and normal breast tissues. One fundamental assumption is that the breast is predominantly characterized by dielectric homogeneity, with notable contrasts in dielectric properties between normal and malignant breast tissues. An integral component of signal processing in any breast imaging system employing MCI is the image reconstruction algorithm. This algorithm plays a pivotal role in determining the efficacy and accuracy of MCI in detecting breast cancer. The choice and refinement of this algorithm are crucial factors in ensuring the precision and reliability of the breast tumor detection process using MCI technology [38, 39].

This approach is based on several fundamental steps for reconstructing a 2D image of the tumor, relying on a total of 5 key stages:

- Data collection
  In this section, we collected data from Sn1 for various antenna positions across a 100 mm by 100 mm area to cover all regions of the breast with a tumor positioned in the middle of the breast. The data collection procedure was repeated for each antenna position for both artificial breasts (with and without tumors) to generate a complete breast image. The transmission coefficient signals collected are S_{XY} and ST_{XY} for the breast without and with tumor, respectively.
- IFFT
  The frequency domain signals are converted into time domain signals by utilizing the Inverse Fast Fourier Transform (IFFT), as illustrated in Eqs. 2 and 3 [38, 39].

$$S_{XY}(t) = IFFT(S_{XY}) \tag{2}$$

$$ST_{XY}(t) = IFFT(ST_{XY}) \tag{3}$$

- Calibration
  The calibration process involves subtracting the response from the tumor-present breast environment from that of the tumor-free breast phantom to eliminate the signals originating from the tumor-free breast environment. This subtraction results in a heightened peak at the tumor location by diminishing the response from the background environment of the target signal as shown in Eq. 4 [38, 39].

$$CT_{XY}(t) = ST_{XY}(t) - S_{XY}(t) \tag{4}$$

- Clutter removal
  The $CT_{XY}(t)$ signal still contain reflections predominantly from the antenna and the environment. To mitigate reflections stemming from the antenna and environment, signal averaging is necessary. Averaging is performed by summing the breast phantom signals and dividing them by the total number of antenna positions, denoted as N. These resulting signals are referred to as averaged signals as illustrated in Eqs. 5 and 6 [38, 39].

$$M_X(t) = \frac{\sum_{Y=1}^{N} CT_{XY}(t)}{N} \tag{5}$$

$$SP_{XY}(t) = CT_{XY}(t) - M_X(t) \tag{6}$$

- Synthetic focusing
  We begin by selecting a region of 300 * 300 (mm) and calculate the distances between each antenna position (XY) and individual pixels (xi and yj) within this area. In our system microwave imaging setup, the distance from the transmitter to a pixel and from that pixel to the receiver is identical. As a result, we double the calculated distance D_{XY} (xi, yj), as shown in Eq. 7. Then, we proceed to calculate the round-trip time it takes for the signal to return to the antenna, as outlined in Eq. 8 [38, 39].

$$D_{XY}\left(x_i, y_i\right) = 2\sqrt{(X - xi)^2 + (Y - yi)^2 + h^2} \tag{7}$$

$$t_{XY}\left(x_i, y_i\right) = \frac{D_{XY}(x_i, y_i)}{C/\sqrt{\varepsilon_r}} \tag{8}$$

- Creating image

  We determine the intensity value I (xi, yj) for each pixel point by examining the signal value in the processed signal $SP_{XY}(t)$ at the specific time t_{XY} (xi, yj), as described in Eq. 9. In this context, 'r' stands for the total number of antenna positions arranged in a row, 's' signifies the overall quantity of antenna positions organized in a column [38, 39].

$$I\left(x_i, y_i\right) = \left[\sum_{X=1}^{r}\sum_{Y=1}^{s} SP_{XY}(t_{XY}\left(x_i, y_i\right))\right]^4 \tag{9}$$

Based on the results obtained during our experiment, it became evident that the proposed microwave confocal imaging algorithm proves to be an effective method for tumor detection. Indeed, upon analyzing the images, we observed that the tumor exhibited a higher intensity compared to other tissue layers, enabling its detection and localization within the breast. The tumor in the image is indicated by the most intense color (a reddish-orange spot). In any case, our experiment demonstrated that microwave confocal imaging is a promising technique for early tumor detection, and future technological advancements are likely to further enhance its performance. This conclusion was derived from a meticulous analysis of the data collected during our study, highlighting the high precision of our system in tumor detection (Fig. 41).

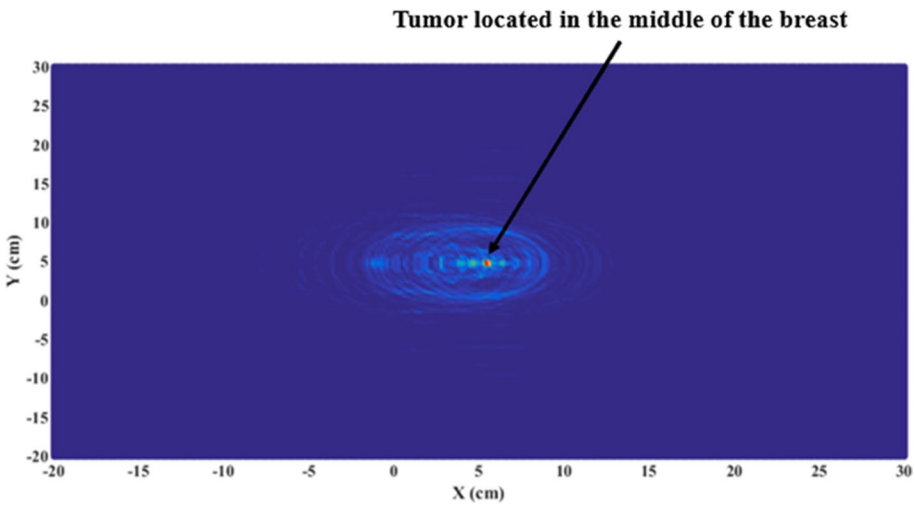

Fig. 41 2D Image of the tumor

9 Conclusion

In this paper, we investigated the possibilities offered by microwaves to achieve a compact imaging and detection system for breast cancer at low cost and without side effects. Several experiments were conducted to evaluate the performance of the antenna array system realized, the results showed that the proposed system could correctly recognize and detect tumors inside a physical mammary phantom performed. A UWB antenna was built covering the entire FCC band from 3.1 GHz to 11.6 GHz. We position a total of 12 antennas in a circular arrangement around the phantom. This antenna array within the imaging system serves as a sensor for transmitting and receiving microwave signals to and from the breast. The signals reflected from the breast are received by the custom-designed antenna and are then subjected to analysis using an appropriate algorithm for tumor detection and localization.

Our future work aims to improve the performance of microwave imaging by developing an algorithm using artificial intelligence that can generate 3D images of breast tumors with high accuracy and precision. The algorithm will be designed to automatically process the data collected from the microwave antenna system and generate 3D images of the breast tissue, including the tumor. The 3D images generated by the algorithm will be of high quality and resolution, enabling clinicians to accurately locate and characterize breast tumors. The system will be optimized to be low-cost and comfortable, ensuring that it can be used in clinical settings and accessible to patients of all socio-economic backgrounds. Additionally, the system will be non-ionizing, reducing the risk of radiation exposure for patients. The development of this technology will be a significant step forward in breast cancer diagnosis and treatment, enabling earlier detection and more effective management of the disease.

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Declarations

Conflict of interest Authors declare no conflict of Interest.

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